TRINAY

AI-ASSISTED RETINAL SCREENING REPORT

CONFIDENTIAL MEDICAL DOCUMENT

DATE: 07 Oct 2026

DEPARTMENT OF TELE-OPHTHALMOLOGY & DIABETIC RETINOPATHY SCREENING

1. PATIENT INFORMATION & SCREENING DETAILS

Patient Name:	**Patient ID:**	**Age / Gender:**	**Patient Phone:**
Walk-in Patient	**PX-3075**	**55 Yrs / Other**	**N/A**
Screening Date:	Examined Eye:	Screening Nurse:	Diabetes Duration:
07 Oct 2026	**Right Eye**	**avi**	**N/A**

2. IMAGE INFORMATION & QUALITY ASSESSMENT

Image Filename: **0b121d798018.png** Image Quality: **Unsuitable** Quality Status: **Diagnostic Grade**

3. AI SCREENING RESULT & CLASSIFICATION

Moderate Non-Proliferative DR (Level 2)

Deep neural ensemble prediction calibrated for primary tele-screening.

CONFIDENCE SCORE

89.7%

Calibrated Model Output

4. PATIENT-SPECIFIC AI CLINICAL SUMMARY

4 retinal lesion types were detected in this screening, including Microaneurysm, Retinal Hemorrhage, Hard Exudate, Soft Exudate (Cotton-Wool Spot), encompassing a total quantified lesion area of 4,948 px. These findings represent retinal abnormalities that may be associated with diabetic retinal vascular disease. Validated neural classifier staging indicates Moderate Non-Proliferative DR (Level 2) (89.7% confidence). Pixel measurements reflect localized lesion load and must be correlated with clinical slit-lamp findings. In-person ophthalmologist review and dilated biomicroscopy are recommended.

Total detected lesion footprint: 4,948 px • Severity staging is derived strictly from the calibrated deep neural ensemble and not lesion pixel area alone.

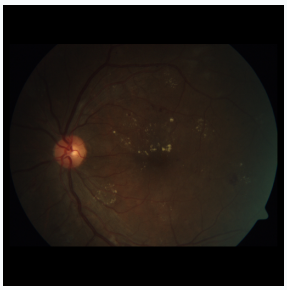
5A. AI-DETECTED FINDINGS (MEASUREMENTS)

Microaneurysm	**Detected (11 px)**
Retinal Hemorrhage	**Detected (3,854 px)**
Hard Exudate	**Detected (382 px)**
Soft Exudate (Cotton-Wool Spot)	**Detected (701 px)**

5B. RETINAL ANATOMY LOCALIZATION

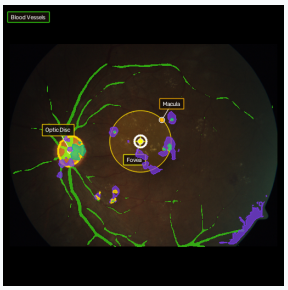
Optic Disc	**1,945 px (Localized)**
Macula / Fovea	**Pos: (249.87, 248.09)**
Retinal Blood Vessels	**6,261 px**
Media Clarity	**Unsuitable**

6. RETINAL VISUAL EVIDENCE GALLERY (UNCROPPED FUNDUS PROPORTION)



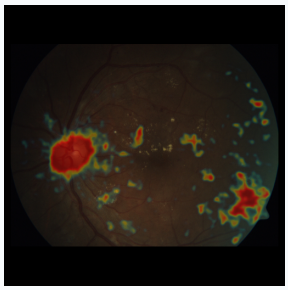
1. Original Fundus

True color clinical capture



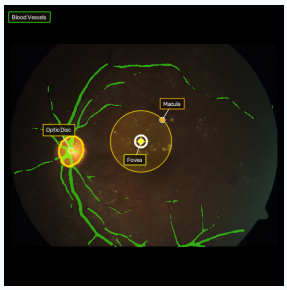
2. Combined Retinal AI

Landmarks & detected lesions



3. Grad-CAM++ Attention

Neural activation heatmap



4. Optic Disc & Vessels

Anatomical reference landmarks

7. AI-DETECTED FINDINGS — DOCTOR-FRIENDLY CLINICAL EXPLANATIONS

Microaneurysm — Detected: 11 px

What it is: A small localized outpouching of a retinal capillary wall where weakened pericytes allow dilation.

Why it matters: Microaneurysms represent the earliest observable retinal microvascular finding associated with diabetic retinopathy.

Clinical Note: The finding should be interpreted together with other retinal abnormalities and the patient's clinical glycemic history.

Retinal Hemorrhage — Detected: 3,854 px

What it is: Bleeding caused by leakage from damaged and ruptured retinal microvessels into intraretinal layers.

Why it matters: Retinal hemorrhages can occur in diabetic retinopathy and other retinal vascular disorders, indicating microvascular compromise.

Clinical Note: Requires clinical correlation and ophthalmologist assessment to evaluate the extent of vascular compromise.

Hard Exudate — Detected: 382 px

What it is: Lipid and lipoprotein deposits precipitated in the retina from abnormal vascular permeability and serum leakage.

Why it matters: They indicate chronic retinal vascular leakage and are important when assessing diabetic retinal disease and proximity to the fovea.

Clinical Note: Evaluate spatial relationship to the foveal avascular zone to monitor risk of diabetic macular edema (DME).

Soft Exudate (Cotton-Wool Spot) — Detected: 701 px

What it is: A localized area of retinal nerve-fiber-layer ischemic injury with axoplasmic flow stasis.

Why it matters: It represents focal retinal ischemic change and can occur in diabetic retinopathy as well as other systemic microvascular conditions.

Clinical Note: Reflects localized precapillary arteriolar occlusion and localized retinal ischemia rather than true exudation.

8. CLINICAL INTERPRETATION & DOCTOR'S TRIAGE SUMMARY

Retinal screening examination reveals findings consistent with Moderate Non-Proliferative DR (Level 2). Visible vascular anomalies detected in retinal vasculature. Recommendation: Vitreo-retinal specialist consultation for dilated biomicroscopy and optical coherence tomography (OCT) to rule out diabetic macular edema (DME).

What this means for the patient: The scan detected signs of diabetic eye changes that need attention from an eye specialist. Early detection allows gentle, effective treatments to protect your eyesight.

9. REFERRAL RECOMMENDATION & STATUS

Referral Status:	Priority Level:	Target Facility:
Referred to Specialist	**Medium Priority**	**Regional Eye Care Center - Vitreo-Retina**

Referral Reason: **Diabetic retinopathy findings requiring clinical slit-lamp dilated examination.**

10. SPECIALIST REVIEW & ASSESSMENT

Specialist Review: Pending

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Specialist Assessment: *Pending specialist clinical evaluation.*

Specialist Notes: *This screening record is queued for tele-ophthalmology verification.*

Specialist Recommendation: *Pending clinical review by assigned specialist.*

11. FOLLOW-UP RECOMMENDATION

Comprehensive dilated ophthalmology evaluation within 2 to 4 weeks. Continue blood glucose and blood pressure monitoring.

12. OFFICIAL REGULATORY DISCLAIMER

TRINAY provides AI-assisted screening support and does not replace diagnosis by a qualified eye-care professional. Definitive clinical decisions should always be confirmed through direct dilated slit-lamp biomicroscopy or OCT by an ophthalmologist.